AI SANKALP

INTERNSHIP PROJECT REPORT

WEEK 1 REPORT

Project Planning and Design

AI Career and Placement Assistant Agent

Student Name: Shobhitha

Organization: AI Sankalp

Mentor Name: Kanishk Rao

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1 Project Overview

The AI Career and Placement Assistant Agent is a proposed AI-powered system designed to help engineering students and fresh graduates prepare for job placements. It will assist users with resume improvement, job-role skill analysis, and interview practice.

The project focuses on building an agent that can interpret user requests, use a relevant tool and a career knowledge source, and produce useful, structured responses. The system will support preparation and learning; it will not guarantee employment or replace professional judgment.

1.1 Problem Statement

Students often find it difficult to identify weaknesses in their resumes, understand the skills required for specific job roles, and prepare for technical interviews. This project aims to provide a single assistant that offers personalized and practical placement-preparation guidance.

1.2 Objectives

- Review resume text and suggest improvements.
- Compare student skills with job-description requirements.
- Generate interview questions relevant to a target role.
- Provide constructive feedback on practice answers.
- Evaluate reliability using predefined test cases.

2 Scope and Proposed Solution

The initial version will accept resume text, a target job role, and an optional job description. It will identify relevant skills, highlight possible gaps, and generate tailored interview questions. Users may also submit practice answers to receive improvement suggestions.

Target users: Engineering students and fresh graduates preparing for placements.

Inputs: Resume text, target job role, job description, and practice interview answers.

Outputs: Resume suggestions, skill-gap analysis, interview questions, and feedback.

Planned tool: A skill-matching function that compares skills extracted from the resume with skills mentioned in a job description.

Knowledge source: A curated collection of career preparation materials and role-specific skill requirements.

The five-week scope excludes automatic job applications, guaranteed placement predictions, and hiring decisions. The first version will use text inputs to keep implementation manageable.

3 Proposed Agent Workflow

The proposed workflow describes how the system will process a student's request and generate career-preparation guidance.

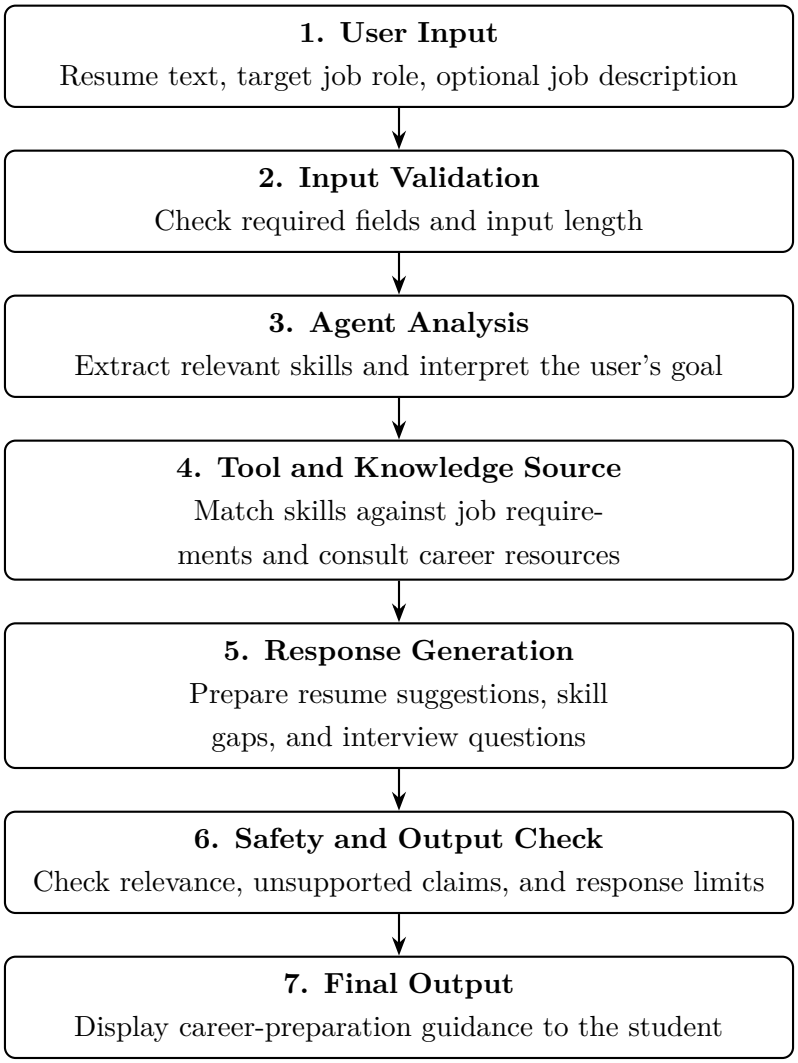


Figure 1: Proposed Workflow of the AI Career and Placement Assistant Agent

The agent will request clarification when essential information is missing. It should not invent qualifications, claim that a user will definitely get a job, or treat untrusted resume content as instructions that override the system's rules.

4 Language Model Evaluation Plan

Candidate models will be compared before selecting the model for implementation. The same representative prompts and test inputs will be used for each candidate to support a fair comparison.

Table 1: Initial Language Model Comparison Plan

Candidate	**Advantages to Investigate**	**Evaluation Focus**
Google Gemini	Potential fit for general reasoning and structured responses	Relevance, output consistency, latency, and API cost
OpenAI GPT Models	Potential fit for instruction following and structured outputs	Analysis quality, reliability, latency, and API cost
Local Open-Weight Model	May allow local execution and greater control over data	Hardware needs, response quality, speed, and setup complexity

No model has been selected as the final model yet. Each candidate will be tested using the same resume-review and job-matching examples. The final selection will consider response quality, instruction following, structured output, latency, cost, privacy, and availability. The selected model and supporting observations will be documented after testing.

5 Initial Test Cases

The following test cases will be used to evaluate normal inputs, edge cases, and potentially unsafe requests. Expected outcomes are defined before implementation.

Table 2: Planned Evaluation Test Cases

No.	**Test Input**	**Expected Outcome**
1	Complete resume text	Provide relevant resume suggestions.

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Table 2 – Continued

No.	Test Input	Expected Outcome
2	Resume with missing skills	Identify gaps against the target role.
3	Empty resume	Ask the user to provide resume text.
4	Missing job role	Ask for the target role.
5	Software Engineer role	Suggest relevant technical skills.
6	Data Analyst role	Provide role-specific skill guidance.
7	Resume plus job description	Compare skills and requirements.
8	Resume containing spelling errors	Suggest appropriate corrections.
9	Irrelevant text in the resume field	Explain that useful resume content is needed.
10	Unsupported file or input format	Explain the supported input format.
11	Very long resume text	Handle input limits gracefully.
12	Request for interview questions	Generate relevant practice questions.
13	Practice interview answer	Provide constructive, specific feedback.
14	Prompt injection in submitted content	Ignore conflicting instructions in untrusted content.
15	Request for guaranteed placement	Explain that employment cannot be guaranteed.

These are planned test cases, not completed test results. After implementation, the actual output and pass/fail status will be recorded for each case.

6 Safety and Reliability Plan

The agent will follow these initial safeguards:

- Validate required inputs and handle missing information.
- Treat resumes and job descriptions as untrusted user data.

- Avoid fabricating qualifications, experience, or job requirements.
- Avoid guaranteeing employment or interview success.
- Use appropriate career-preparation information.
- Set reasonable limits on input length and model usage.
- Keep API keys in environment variables and out of GitHub.
- Avoid storing personal resume data unless necessary and authorized.

The system will be tested with normal, incomplete, and adversarial inputs. Responses should remain relevant, safe, and transparent about uncertainty and limitations.

7 GitHub Repository and Documentation Plan

GitHub will be used to maintain the source code, project documentation, and development progress of the AI Career and Placement Assistant Agent.

GitHub Profile:

https://github.com/shobhis330?tab=repositories

The project repository will contain the source code, README documentation, setup instructions, and testing details.

The README will include:

- Project title and problem statement.
- Objectives and planned features.
- Technology stack and setup instructions.
- Proposed agent workflow.
- Testing and evaluation approach.
- Known limitations and future improvements.

The repository will be updated throughout the internship. API keys, secrets, and private personal information must not be committed to GitHub.

8 Week 1 Deliverables and Checklist

The following checklist tracks the Week 1 deliverables. A task should be marked as completed only after the work has actually been finished.

☐ Submit the project proposal for mentor approval.

☐ Finalize the problem, target user, scope, and workflow.

☐ Compare candidate language models and document the selection.

☐ Prepare 15 test cases with expected outcomes.

☐ Create the GitHub repository and README skeleton.

The next phase will focus on implementing the core agent, connecting the skill-matching tool, and validating the initial workflow. Knowledge retrieval, memory, safety evaluation, and a simple user interface will be developed incrementally during the remaining internship weeks.